Soohyun Choi

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning
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Research Interests

My research interests lie in reinforcement learning, particularly in the mathematical and geometric analysis of learning algorithms. I am also interested in developing new algorithms and modeling frameworks using generative models and stochastic processes, especially for goal-conditioned and long-horizon control.

Education

Hanyang University — Sep. 2024–Present

Integrated M.S./Ph.D. Program in Electronic Engineering
Information and Intelligence Systems Lab (IISL) • Advisor: Prof. Songnam Hong

Hanyang University — 2018–2024

B.S. in Electronic Engineering • Graduation Honors

Manuscripts & Research Projects

PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning — 2026

Soohyun Choi, Seonvin Cho*, and Songnam Hong • arXiv preprint • *Equal contribution • Code • arXiv*
Developed a hierarchical offline GCRL framework that connects subgoal selection to short-horizon execution through state-space bridges, inverse dynamics, and execution-time replanning.

Multi-step Proximal Policy Improvement in Offline Reinforcement Learning — 2026

Soohyun Choi, Seonvin Cho*, and Songnam Hong† • arXiv preprint • *Equal contribution • †Corresponding author • Code*
Developed a plug-in offline policy-refinement framework that composes re-centered proximal steps on a policy manifold and analyzes how step size and critic error govern finite-step improvement.

ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian Manifold — 2024; rev. 2026

Soohyun Choi, Minseok Oh, and Ickhyun Song • Undergraduate capstone technical report • Report
Formulated designer-selected performance equivalence classes and quotient geometry for circuit search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

Ongoing Research

MART: Multi-Step Actor Refinement Trajectories in Offline Reinforcement Learning — 2026–Present

Study how dividing a fixed actor-improvement horizon across persistent local actors changes the stability frontier, critic-facing bootstrap exposure, and deployment reach in behavior-regularized offline RL.

AMO: Adaptive Multiscale Policy Optimization in Offline Reinforcement Learning — 2026–Present

Learn the total policy-improvement horizon T through an offline bilevel outer objective while treating N as a multiscale resolution parameter; the critic uses the local T/N policy, whereas the actor represents the full-horizon T policy.

PathFlower: Goal-Conditioned State Flows — Ongoing

Extend PathBridger toward final-goal-conditioned generative state flows that produce useful intermediate states for long-horizon offline control.

Honors & Funding

Top Excellence Award - Alumni Association Special Award (KRW 1,000,000) — 2023

Top Excellence Award - 15th Capstone Design Fair (KRW 1,000,000) — 2023

Graduation Honors, Hanyang University — 2024

BK Research Assistantship, Hanyang University — Fall 2025

Selected Graduate Coursework

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory
Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I